Phase 1 Implementation Blueprint

Project Title: Precision Cardiology: A Multi-Modal Digital Twin for 24-Hour Acute Ischemic Event Forecasting and Hemodynamic Visualization

1. Strategic Objective

The objective of this project is to build an advanced predictive Digital Twin proof-of-concept that successfully merges static clinical structures with real-time dynamic biomarkers. This tool allows clinicians to foresee sudden ischemic cardiac events up to 24 hours before onset and interact with a real-time, high-fidelity 3D model of the patient's hemodynamic system.

2. Data Architecture & Privacy Compliance

To remain completely compliant with data privacy frameworks such as the **DPDP Act** and **HIPAA**, the prototype operates entirely within a sandboxed environment using synthetic and fully anonymized open-source data streams.

A. Static Historical Stream (EHR Profile)

- **Source:** Generated using the Synthea Patient Generator ecosystem or curated from the MIMIC-IV Clinical Database.
- **Core Features:** Patient Demographics (Age, Biological Sex, BMI), Comorbidities (Hypertension, Hyperlipidemia, Type-2 Diabetes), Clinical History (Previous ischemic episodes, smoking index), and Baseline Diagnostics (LVEF %, Resting ECG metrics).

B. Dynamic Real-Time Stream (Wearable/IoT Sensors)

- **Source:** Extracted from the open-access PhysioNet Multi-parameter Databases or synthetic time-series generation algorithms.
- **Core Features:** Continuous heart rate variability metrics (HRV RMSSD, SDNN), Sleep state architecture data (Deep, REM, and Light sleep fragmentation indices), and Activity loads (Active Energy Expenditure, step frequency).

3. Predictive Algorithmic Engine

The modeling infrastructure comprises a multi-modal neural network or ensemble gradient booster (e.g., XGBoost / LightGBM) that reads historical structural risks alongside sliding windows of biometric time-series data.

Data Stream Type	Frequency	Analytical Value

Static EHR Matrices	Baseline / Static	Establishes prior conditional risk weights and physiological boundaries.
Dynamic Wearable Vectors	Continuous / Intraday	Tracks acute shifts, autonomic nervous system stress, and sleep degradation.

4. Clinical UI Dashboard Strategy

The interface functions as a web-accessible clinical cockpit, separating detailed tabular parameters from high-impact visual computing tools.

- **Clinical Command Panel:** Hosts standard EHR diagnostic summaries alongside real-time time-series telemetry widgets.
- **Forecasting Metrics Banner:** Explicitly showcases a forward-looking risk coefficient (e.g., *'Acute Ischemic Anomaly Risk: 84% [Window: 18-24h]'*) alongside key dynamic triggers.
- **The Hemodynamic 3D Engine:** A canvas element powered by WebGL/Three.js animating a human heart model. Fluid dynamics are represented using chromatic shifts. Normal vascular parameters animate stable blue/red gradients; imminent predictive risk spikes cause targeted vascular cross-sections to pulse in amber/red.

5. Tactical Action Plan

Milestone	Tasks & Technical Requirements
Step 1: Environment Setup	Initialize clean GitHub repo with structured paths (`/src`, `/notebooks`, `/dashboard`). Draft core dependencies file (`requirements.txt`).
Step 2: Data Hydration	Execute Python scripts synthesizing combined clinical-wearable datasets. Bind both tables together using unique pseudo-patient tokens.
Step 3: Model Training	Build feature matrices combining rolling window HRV values with fixed risks. Train baseline predictive classifiers and export metrics.
Step 4: UI Concept Build	Design clear Figma interaction mockups or engineer an active, easy-to-use Python Streamlit page to host interactive 3D assets.

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